

Bucket Versioning.

What it actually is

Bucket Versioning in **Amazon S3** means every time you upload, modify, or delete a file (object), S3 keeps the old version instead of overwriting it.

No versioning = new upload replaces old file permanently.

With versioning = old file stays, new file gets a unique **Version ID**.

Why It Matters (Real Use Cases)

1. Accidental delete recovery

- If someone deletes project.zip, S3 just adds a **Delete Marker**.
- Old versions still exist.
- You can restore them.

2. Rollback capability

- Broke production config?
- Just restore previous version.

3. Protection from overwrites

- If someone uploads wrong data, you don't lose original.

Bucket versioning is only suspended not to be disabled , and I have enabled after creating the bucket.

Edit Bucket Versioning [Info](#)

Bucket Versioning

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning

☐ Suspend

This suspends the creation of object versions for all operations but preserves any existing object versions.

☒ Enable

ⓘ After enabling Bucket Versioning, you might need to update your lifecycle rules to manage previous versions of objects.

Multi-factor authentication (MFA) delete

An additional layer of security that requires multi-factor authentication for changing Bucket Versioning settings and permanently deleting object versions. To modify MFA delete settings, use the AWS CLI, AWS SDK, or the Amazon S3 REST API. [Learn more](#)

Disabled

[Cancel](#)

[Save changes](#)

Bucket versioning works like an commit id in github.

I have uploaded 3 files to the bucket.

Amazon S3

Upload: status

Close

After you navigate away from this page, the following information is no longer available.

Summary

Destination

s3://bucketversioning34565

Succeeded

3 files, 5.0 MB (100.00%)

Failed

0 files, 0 B (0%)

Files and folders

Configuration

Files and folders (3 total, 5.0 MB)

Find by name

Name	Folder	Type	Size	Status	Error
Day 2 Linux Navigation Comma...	-	application/pdf	1.4 MB	Succeeded	-
Day 3 admin and permission.pdf	-	application/pdf	1.9 MB	Succeeded	-
Day 4 networking.pdf	-	application/pdf	1.6 MB	Succeeded	-

The uploaded files.

bucketversioning34565

Info

Objects

Metadata

Properties

Permissions

Metrics

Management

Access Points

Objects (3)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Show versions

Name	Type	Version ID	Last modified	Size	Storage class
Day 2 Linux Navigation Commands.pdf	pdf	GK5.ThDKPDRuvfwfDvDg8fCDU96GWEPE	February 20, 2026, 17:14:43 (UTC+05:30)	1.4 MB	Standard
Day 3 admin and permission.pdf	pdf	_jWtSierJODXWismh4duFegLwaH09CRJ	February 20, 2026, 17:14:44 (UTC+05:30)	1.9 MB	Standard
Day 4 networking.pdf	pdf	.dJnErravNT4n3lw527yal35N5C rp_BH	February 20, 2026, 17:14:45 (UTC+05:30)	1.6 MB	Standard

The bucket versioning enabled means the bucket will create and version id we can retriever the older version when we delete the file.

newbucket4622982

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Name	Type	Version ID	Last modified	Size	Storage class
images/	Folder	-	-	-	-
index.html	Delete marker	tLp2l0jvt3fZlYnedCdKh3zLSpGay	February 20, 2026, 17:04:34 (UTC+05:30)	0 B	-
index.html	html	V7G9HfcUFQXSiqhLQVuiVTaC.LVQBZC	February 20, 2026, 16:57:58 (UTC+05:30)	18.9 KB	Standard
templatemo-606-string-master.css	Delete marker	1SjneELBL6c7PK1LJBpN_9SZe uWW6bZ	February 20, 2026, 17:04:34 (UTC+05:30)	0 B	-
templatemo-606-string-master.css	css	1rqk73MnWN3cmQ6JJwT80VaOxNCuB0u	February 20, 2026, 16:57:58 (UTC+05:30)	23.1 KB	Standard
templatemo-606-string-scripts.js	Delete marker	BjSijoOlx0zLxfWZy5h2qeKNcFFyJwsa	February 20, 2026, 17:04:34 (UTC+05:30)	0 B	-
templatemo-606-string-scripts.js	js	IcEn4rmDmiL4lwAq4e0XgyJ0WcBIOOx	February 20, 2026, 16:57:59 (UTC+05:30)	13.2 KB	Standard

Bucket Replication.

What It Actually Is

S3 Bucket Replication automatically copies objects from one S3 bucket to another.

There are two types:

- **CRR (Cross-Region Replication)** → Different AWS regions
- **SRR (Same-Region Replication)** → Same region

Why Replication Exists

Disaster Recovery

- If your primary region fails, your data exists elsewhere.

Low Latency Access

- Users in another country? Store data closer to them.

Compliance

- Some laws require data to exist in specific regions.

Multi-Account Strategy

- Replicate from production account → backup account.

TYPES OF REPLICATION.

1. Cross-Region Replication (CRR)

Replicates objects to a **different AWS region**.

Example:

- Source → Mumbai region
- Destination → Singapore region

Used for:

- Disaster Recovery
- Global applications
- Region-level failure protection

Reality:

- Costs more (data transfer charges apply)
- Good for serious production setups

2. Same-Region Replication (SRR)

Replicates objects to another bucket in the **same region**.

Used for:

- Log copying
- Compliance separation
- Multi-team access segregation

Cheaper than CRR (no cross-region transfer cost)

3. Cross-Account Replication

Replication between **different AWS accounts**.

Used for:

- Production account → Backup account
- Security isolation
- Enterprise setups

Requires:

- IAM role permissions

- Bucket policy on destination

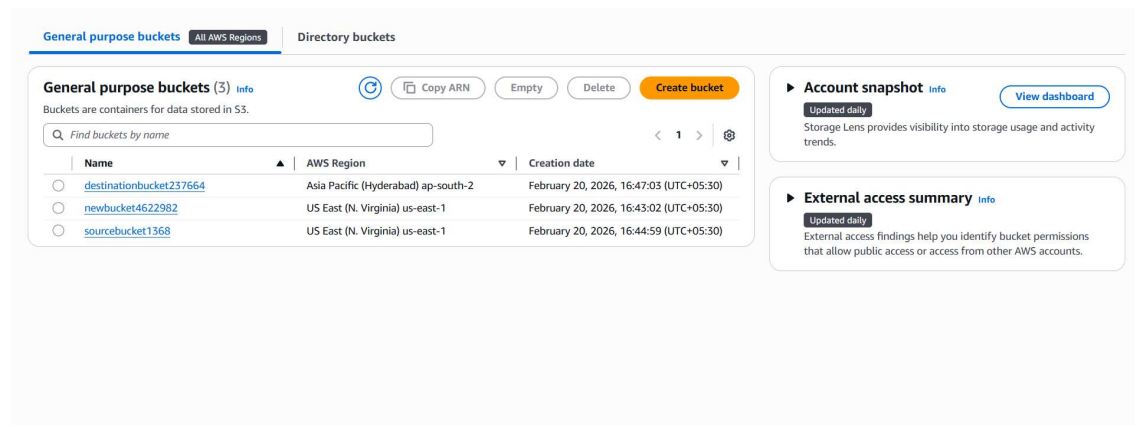
This is how mature DevOps teams operate.

4. Delete Marker Replication

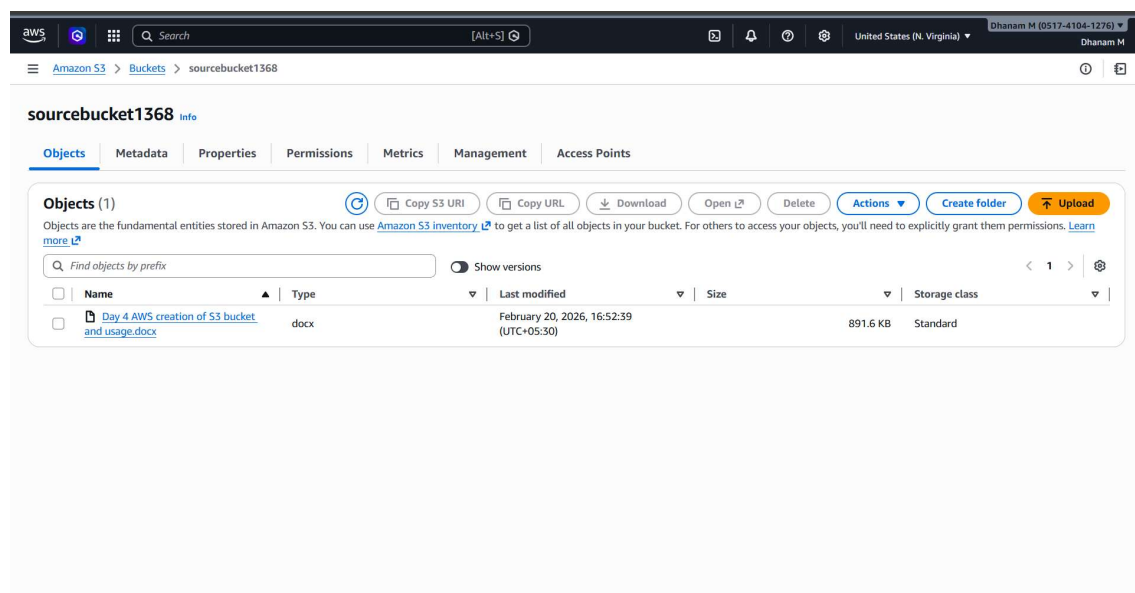
Controls whether object deletions are replicated.

- Enabled → Delete in source = Delete in destination
- Disabled → Destination keeps object even if source deletes

I have created two bucket in different regions.



This the source bucket.



Create replication rule.

Amazon S3 > Buckets > amazonsourcebucket3457 > Replication rules > Create replication rule

Replication rule name
temp_rule
Up to 255 characters. In order to be able to use CloudWatch metrics to monitor the progress of your replication rule, the replication rule name must only contain English characters.

Status
Choose whether the rule will be enabled or disabled when created.
☒ Enabled
☐ Disabled

Priority
The priority value resolves conflicts that occur when an object is eligible for replication under multiple rules to the same destination. The rule is added to the configuration at the highest priority and the priority can be changed on the replication rules table.
0

Giving the destination bucket details.

Amazon S3 > Buckets > amazonsourcebucket3457 > Replication rules > Create replication rule

☐ Limit the scope of this rule using one or more filters
☒ Apply to all objects in the bucket

Destination
Destination
You can replicate objects across buckets in different AWS Regions (Cross-Region Replication) or you can replicate objects across buckets in the same AWS Region (Same-Region Replication). You can also specify a different bucket for each rule in the configuration. [Learn more](#) or see [Amazon S3 pricing](#)
☒ Choose a bucket in this account
☐ Specify a bucket in another account

Bucket name
Choose the bucket that will receive replicated objects.
amazondstibucket32566 [Browse S3](#)

Destination Region
US East (N. Virginia) us-east-1

Create own IAM role.

Amazon S3 > Buckets > amazonsourcebucket3457 > Replication rules > Create replication rule

Destination Region
US East (N. Virginia) us-east-1

IAM role
Permission to access the specified resources
☒ Create new role
☐ Choose from existing IAM roles
☐ Enter IAM role ARN

If you want the existing data tick the yes option.

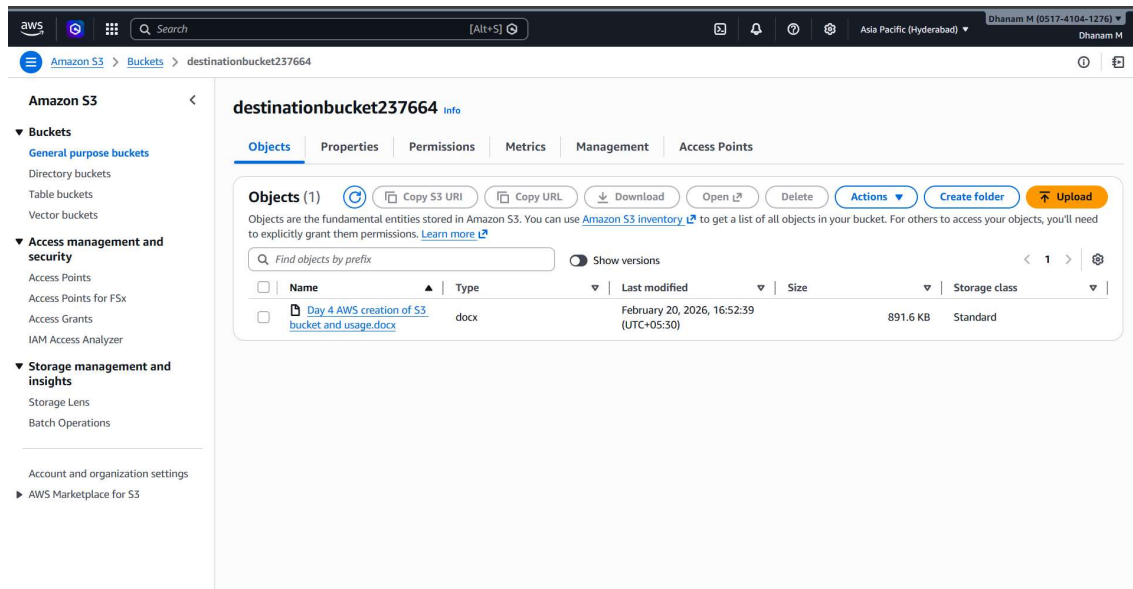
Replicate existing objects?

You can enable a one-time Batch Operations job from this replication configuration to replicate objects that already exist in the bucket and to synchronize the source and destination buckets. [Learn more](#) or [see pricing](#)

Existing objects
☐ No, do not replicate existing objects.
☒ Yes, replicate existing objects.

[Cancel](#) [Submit](#)

The destination bucket will be updated from 15 minutes to max hours.



Static Web Hosting.

What You Actually Get

- Cheap hosting
- High durability (11 9's)
- Auto scaling
- Public HTTP endpoint
- No server management

What you don't get:

- No server-side code execution
- No HTTPS by default (you need CloudFront for that)
- No database

To enable the static web hosting in the s3 we have to enable ACLs , Bucket versioning , and allow all public access (I have already done it).

aws

Search

[Alt+S] Ask Amazon Q

United States (N. Virginia)

Dhanam M

Amazon S3

Buckets

newbucket4622982

newbucket4622982

info

Objects

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☐	templatemo-606-string-scripts.js	js	February 20, 2026, 16:57:59 (UTC+05:30)	13.2 KB	Standard

ebhosting8745467

Successfully edited static website hosting.

Requester pays

Disabled

Static website hosting

Edit

Use this bucket to host a website or redirect requests. Learn more

We recommend using AWS Amplify Hosting for static website hosting

Deploy a fast, secure, and reliable website quickly with AWS Amplify Hosting. Learn more about Amplify Hosting or View your existing Amplify apps

Create Amplify app

S3 static website hosting

Enabled

Hosting type

Bucket hosting

Bucket website endpoint

When you configure your bucket as a static website, the website is available at the AWS Region-specific website endpoint of the bucket. Learn more

http://staticwebhosting8745467.s3-website-us-east-1.amazonaws.com

The hosted website is, using the given s3 url.

